

Yuchen Liu

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AI systems engineer building RL training infrastructure, search/data pipelines, and research-grade open-source tools.

EDUCATION

M.S. Computer Science	University of Pennsylvania, Philadelphia	GPA: 3.80/4.00	May 2025
B.S. Applied Economics	Pennsylvania State University, State College	GPA: 3.86/4.00	May 2023

WORK EXPERIENCE

Software Engineer, Search Infrastructure | DoorDash Aug 2025 - Present | Seattle, WA

- Built Notus, a next-generation search ingestion platform using Iceberg, Kafka, and Spark to decouple document ingestion from serving, enable configuration-driven indexing, and unblock schema evolution for hybrid search.
- Built Vela, Argo's hybrid retrieval platform, enabling Lucene-native hybrid search and federated retrieval with vector backends such as Milvus behind a single API.
- Migrated 30+ production stacks to modular Jsonnet, eliminating roughly 56,000 lines of duplicate configuration and reducing config verification time by 90%.
- Stabilized production indexers with 500GB EBS storage, pre-deployment validators, and Kyverno guardrails after RCA for \$21K revenue-risk incidents, preventing capacity-related outages.

Software Engineering Intern | Penn Medicine TissueLab Oct 2024 - May 2025 | Philadelphia, PA

- Developed a Python/FastAPI AI microservice platform to orchestrate PyTorch models for medical image analysis, enabling natural-language-driven segmentation and classification workflows.
- Implemented an event-driven DAG workflow engine for async task execution, processing 10,000+ pathology images daily and streaming 12M+ imaging events through WebSockets with sub-10ms latency.
- Integrated generative and discriminative deep learning models into a unified service layer with asynchronous REST APIs for cross-department usage.

Software Engineering Intern | Information Technology of CAS Apr 2024 - Aug 2024 | Remote

- Designed Kafka and Redis-backed event streaming services that reduced p95 API latency from 2s to 200ms while processing millions of IoT events.
- Contributed to distributed Spring Boot monitoring services, delivered 35+ REST APIs, and maintained 99.7% uptime in Unix-based production environments.

OPEN SOURCE

nanoRL

- Led an open-source RL training framework inside nanochat for clean objective definitions, rollout experiments, RL algorithm research, reproducible ablations, and vLLM inference serving.

nanochat

- Built a hackable pretraining and chat stack supporting architecture definition and FLOP-controlled model ablations.

llm-interp

- Authored reproducible interpretability scripts for model circuit research, including attention-sink analysis and reproductions of LLM decode indeterminism findings.

RESEARCH

Do Value Vectors in Deep Layers Need Context from the Residual Stream?

EMNLP 2026, under review

Co-first author. Challenged the standard attention mechanism of computing value vectors from the residual stream, finding that deep layers benefit from context-free value vectors that preserve original token information.

Proposed Bank of Values, a learned value-vector table for the last third of layers that eliminates the V cache and improves validation loss and average scores across 21 benchmarks on 135M and 780M models.

Vision Language Models Cannot Plan, but Can They Formalize?

ECCV 2026, under review

Studied whether VLMs can formalize visual planning tasks into solver-executable PDDL problem files.

A co-evolving agentic AI system for medical imaging analysis

Nature, under review

Built agentic medical-imaging workflows for tool selection, planning, knowledge updates, and human-in-the-loop co-evolution.

SKILLS

Languages: Python, Kotlin, Java, C++, JavaScript; ML/Infra: PyTorch, vLLM, Spark, Kafka, Iceberg, Delta Lake, FastAPI, Redis, Docker, Kubernetes, AWS, Spring Boot